Christina Saravanos

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Personal Information

Swiss Resident (B permit), EU Citizen, US Citizen

Scientific Interests

Bridging AI, research and innovation into production

• Data Science • Machine Learning • Agentic AI • Natural Language Processing • Signal Processing

Skills

Programming Languages: Python, MATLAB ●●●●● Java, shell ●●●●○
Python Packages: langchain, tensorflow, pytorch, pandas, spaCy, nltk, flask, librosa, opencv, skimage
Version Control: git **NoSQL Databases:** Mongo, Cassandra **Cloud Computing:** Storm, Spark

Experience

Nestlé *February 2024 - Present*
Research Associate, Data Scientist & AI Engineer *Lausanne, Switzerland*

- **AI-Enabled Scientific Intelligence**(Main Project)
  Brief Description: Development of a platform that comprises an AI-enabled conversational interface, an internal search engine that utilizes hybrid search, vector databases, RAG, information extraction via tool-calling and an automated literature review system leveraging advanced Generative and Agentic AI to facilitate researchers in accelerating their research.

  Key Responsibilities:
  * Ownership of the development (designed, implemented and tested) D the AI stream of the project.
  * Adapt and evolve the AI components (of the search engine, chatbot, literature review system) of the project from the Proof-of-Concept phase to the scaling-up and production phases in an agile environment.
  * Fine-tuned the AI components towards various use cases and presented the components, the respective results and findings to potential candidate users.
  * Worked in an international, multi-disciplinary, cross-functional team and collaborated with developers, data engineers and data scientists based in Switzerland, Portugal and India.
  * Continuously tested and evaluated the tasks assigned to the AI stream.

  Key Contributions:
  * Implementation and optimization of the AI components in Python used in various use cases during the different stages of the project's life-cycle.
  * Researched and evaluated the effectiveness of various state-of-the-art models and technologies during the different stages of the project's life-cycle.
  * Transformation of the AI components from prompt-based approaches to agentic workflows to be integrated in autonomous agents.
  * Fine-tuned the LLMs employed in the project using Retrieval Augmented Generation (RAG).

* Applied advanced an agentic reasoning techniques (Chain-of-Thought, tool-use) to implement more complex task, e.g., writing of a report, validiting the references of a response.
* Implemented robust components that are continuously used as the key components of new cases.
* Evaluated the outputs of the pipeline using and combining different approaches and LLMs.
* Implementation of a jupyter notebook to evaluate through basic statistics the user metrics .

Outcomes:

* Several use-cases of the platform were down-selected and moved on to the subsequent phases the project's life-cycle.
* Successfully completed and timely delivered in the AI pipelines of the internal search engine, chatbot and automated literature review process.
* Several use-cases of the platform were down-selected and moved on to the subsequent phases the project's life-cycle.
* The platform started from its conceptualization phase and has successfully scaled-up and is now in its production phase .
* The platform was launched in the summer for internal use and is now being used by approximately 200+ researchers and scientists within Nestlé Research and is expected to have a large impact in Nestlé.

– **Cell Colony Detection** (Additional Project)
Brief Description: Development of a pipeline in Python that detects and counts cell colonies from microscopy images via image processing, computer vision and machine learning techniques.
Key Responsibilities:

* Led the technical aspect, and developed (designed, implemented and evaluated) a pipeline that automatically detects all the cell colonies in microscopy images using filtering, segmentation, object detection, and supervisted and unsupervised learning techniques.
* worked in a multi-disciplinary team that comprised scientists based in Switzerland
* Adapt the pipeline from the Proof-of-Concept phase to the Scaling-up phase of the project.

Key Contributions:

* Implementation of the pipeline during the different stages of the project's life-cycle.
* Conducted research and evaluated the effectiveness of various conventional, state-of-the-art and cutting-edge models and techniques.

Outcomes:

* Achieved a 95% accuracy of detecting cell colonies using conventional computer vision techniques.
* Achieved a 90% accuracy of detecting cell colonies using regression techniques.

– **Automation of Writing The Discoveries of Ongoing and Future Projects.** (Additional Project)
Brief Description: Development of a pipeline in Python to facilitate Project Managers by automating the process of writing the discoveries of projects via Generative AI.
Key Contribution

* Designed, implemented and developed an LLM-based application in Python to create a paragraph with the discoveries of various projects.

Prisma Electronics S.A. *February 2023 - September 2023*
Data Scientist *Athens, Greece*

– **Forecasting A Ship's State Based On Digital Twins and Deep Learning Frameworks** (Main Project)
Brief Description: Development and testing of a pipeline used to forecast the future forecast the future state of ships based on time-series obtained from Digital Twins and Recurrent Neural Networks (RNNs).

– **Estimating Air Pollutant Concentration In Urban Areas** (Additional Project)
Brief Description: Development and testing of a pipeline used to estimate the concentration of air pollutants in Greek cities using data obtained from sensors before and during COVID-19 and regression techniques .

Other Responsibilities:

- ∗ Contributed to writing proposals for soliciting funding by the European Union.
- ∗ Wrote articles and technical reports.

IBM Research *September 2021 - February 2022*
Research Intern, Scalable Knowledge Ingestion *Ruschlikon, Zurich, Switzerland*

– **Python Packages For Entity Recognition And Relationship Extraction From Entity Reports Via Natural Language Processing (NLP)**(Project)

Brief Description: Development and testing of Python packages for Named Entity Recognition and Relationship Extraction annotators via NLP to find and extract entities and their respective relationships from unstructured PDF documents concerning enterprises.

Key Contributions:

- ∗ Developed effective Named Entity Recognition and Relationship Extraction annotators in Python to identify and extract entities and their respective relationships from text and tables of PDF documents.
- ∗ Developed several work flows to populate a Knowledge graph using the entities and the relationships obtained by using the annotators.

Outcomes:

- ∗ Successfully completed and delivered in a timely matter the annotators.
- ∗ The annotators were integrated into IBM's DeepSearch (released in 2022) and by IBM and its clients.
- ∗ The project was presented at the AMLP EPFL 2022 workshop.

SwissRe *November 2020 - May 2021*
Actuarial Control and Claims Reserving Intern *Zurich, Switzerland*

– **Hierarchical Compartmental Reserving** (Project)

Brief Description: Development and testing of a pipepline to predict future cumulative and incremental outstanding and incurred losses via prediction loss models, forecasting algorithms and compartmental reserving models.

Key Contribution

- ∗ Collaborated with the firm's actuaries to develop and implemented in R and evaluated prediction loss models forecasting techniques used for actuarial data e.g., one- and two-stage compartmental reserving models to predict the incurred and outstanding losses using data obtained from market cycles from 2000 to 2019.

The Goodyear Tire & Rubber Company *March 2020 - August 2020*
Data Science Intern *Colmar Berg, Luxembourg*

– **Treadwear Tire Prediction Via Telematic Data And Machine Learning Techniques** (Project)

Brief Description: Development of a pipeline used to predict tire quality and treadwear and identify vehicles using data extracted from Controller-Area-Network (CAN) signals obtained from sensors and Deep Learning (DL) frameworks.

Key Contribution

- ∗ Developed and evaluation of a pipeline in Python that calibrates sensors placed in fleets of cars, predicts future metrics and tire treadwear, identify vehicles using data from CAN signals and RNNs and shows the results in a webpage that ran locally built via Flask.

Outcome

- ∗ The project was integrated into Goodyear's analytics infrastructure and has been used in the U.S. and Europe to calibrate sensors and predict the performance and the wear of tires.
- ∗ Successfully completed the key deliverables of the project.

IBM *April 2019 - July 2019*
Data Science Intern, Defense, NATO and Intelligence team , IBM Benelux *Brussels, Belgium*

- **CV Verification And Augmentation** (Project)

  Brief Description: Development of a pipeline for CV verification that cross-references candidates' CVs by scraping their profiles from LinkedIn and their social media profile pages via NLP and ML techniques.

  Key Contribution

  ∗ Development and testing of a pipeline in Python that scrapes the CVs potential candidates from LinkedIn and verifies them against the content of their social media pages which were extracted by using several crawlers; the results were displayed in a webpages created via Flask.

  Outcome

  ∗ The project was integrated into IBM's overall Curriculum Vitae (CV) Verification platform

  ∗ Successfully completed the key deliverables of the pipeline.

Education

University of Patras, *2020*

M.Sc. in Signal Processing and Communication Systems *Patras, Greece*

GPA: 8.4/10 - *magna cum laude*

Thesis: **"Audio-Fingerprinting Via The K-SVD Algorithm"** *Advisor: Prof. K. Berberidis*

Key Developments:

∗ Proposed, designed, developed and evaluated a novel, robust audio-fingerprinting scheme utilizing sparse coding and dictionary learning techniques.

∗ The proposed audio-fingerprinting scheme proved to be more effective and robust than audio-fingerprinting paradigms that rely on conventional signal processing techniques (e.g., the algorithm applied by Shazam).

∗ The code was implemented in MATLAB.

∗ Presented in the 20th IEEE International Workshop on Multimedia Signal Processing 2020.

University of Patras, *2015*

Engineering Diploma (Integrated M.Sc. and B.Sc.) in Computer Engineering and Informatics *Patras, Greece*

Minor: Software Engineering, Data Science

GPA: 6.81/10 - *suma cum laude*

Thesis: **"Mining Knowledge From Social Networks"** *Advisor: Asoc. Prof. C. Makris*

Key Developments:

∗ Designed, developed and evaluated a pipeline to find X (Twitter)'s most influential users in real time by combining several clustering and voting algorithms which were modified to run for streaming data.

∗ The code was implemented in JAVA, and ran on a Storm cluster, while NoSQL databases were used to dynamically update the parameters of the clustering algorithms.

∗ Presented in the 12th IEEE International Conference on on Information Intelligence Systems and Applications 2021.

Publications

Journal Articles

1. **C. Saravanos** and A. Kanavos, "Predicting stock market alternations using social media sentiment analysis", *Neural Computing and Applications*, 2024.

2. C. Spandonidis, D. Paraskevopoulos, and **C. Saravanos**, "Neighborhood-level particle pollution assessment during the covid-19 pandemic via a novel iot solution", *Sustainability*, 2023.

Conference Proceedings

1. **C. Saravanos** and A. Kanavos, "Forecasting stock market alternations using social media sentiment analysis and regression techniques", in *Artificial Intelligence Applications and Innovations*, Springer.
2. C. Spandonidis, D. Paraskevopoulos, and **C. Saravanos**, "Design and implementation of a digital twin of a ship for sustainable operations", in *17th Ann. Conf. of the Hellenic Marine Technology Institute*.
3. **C. Saravanos** and A. Kanavos, "Forecasting stock market alternations using social media sentiment analysis and deep neural networks", in *14th International Conference on Information, Intelligence, Systems & Applications (IISA)2023*, IEEE, 2023.
4. **C. Saravanos**, G. Drakopoulos, A. Kanavos, E. Kafeza, and C. Makris, "Discovering influential twitter authors via clustering and ranking on apache storm", in *12th International Conference on Information, Intelligence, Systems & Applications*, IEEE, 2021.
5. **C. Saravanos**, D. Ampeliotis, and K. Berberidis, "Audio-fingerprinting via dictionary learning", in *IEEE 22nd International Workshop on Multimedia Signal Processing (MMSP)*, IEEE, 2020.

Projects

- Entity Extraction From Large Document Datasets Via LLMs
- Automated Playlist Generation Via Deep Learning
- Music Genre Classification Via Deep and Dictionary Learning
- Cover Song Identification *(collaborated with C. Theophilou)*
- Song Identification via Data-Driven Dictionaries *(with Prof. D. Amepliotis and Prof. K. Berberidis)*
- Predicting the U.S. Stock Market Volatility Via Social Media Analysis *(with Prof. A. Kanavos)*

Languages

- **English**, Native Lagnuage, *EXAMS: Certificate of Proficiency In English- University of Michigan*
- **French**, Advanced, *EXAMS: Diplome d'Etudes En Langue Francaise (DELF 1er Degre)*
- **Greek**, Native Language,

References

- Peter W.J. Staar,
  Principal Research Staff Member,
  Master Inventor, Manager of
  "AI for Knowledge" group,
  IBM Research Zurich,
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